FINTECH • AI

FinWise AI

Personal Financial Analysis & Smart Budget Assistant

A LangChain + Streamlit educational prototype that combines deterministic Python calculations with structured AI insights.

Presented by: Wajiha

MONTHLY SNAPSHOT

Income

8,000

Expenses

2,000

Remaining

6,000

HEALTH SCORE

82 STRONG

The money problem is often a visibility problem

FinWise starts with the user's actual monthly numbers—not generic advice.

**“Where is my money going,
and what should I improve?”**

The project identifies the need for a clear, structured view of spending and personalized educational guidance.

THE GAP

**Numbers exist.
Clarity is missing.**

01 SPENDING

People may lack a structured view of where their money goes.

02 ADVICE

Generic online advice rarely reflects an individual's actual numbers.

03 PLANNING

Users need practical budgeting suggestions tied to their goals.

FinWise's response

Analyse → Explain → Prioritize → Act

From raw numbers to a practical monthly plan

A hybrid architecture keeps calculations deterministic and AI insights structured.

INPUT

Income • expenses
savings • goal

CALCULATE

Python ratios
+ preliminary score

ANALYSE

LangChain + LLM
structured JSON

ACT

Dashboard +
action plan

The LLM does not replace the math—it turns calculated financial context into educational, personalized guidance.

One dashboard. The whole monthly picture.

Inputs, metrics, AI analysis and next-month actions in one Streamlit experience.

FINANCIAL OVERVIEW

INCOME	EXPENSES	REMAINING	SAVINGS
8,000	2,000	6,000	1,500

FINANCIAL HEALTH

82

Risk: LOW • Goal: Save Money

Smart Inputs

Monthly income + expense categories + savings + financial goal.

AI Analysis

Summary, spending observations, risk, priorities and recommendations.

Live Recommendations

Final recommendations stream into the UI for a natural typing effect.

Built as a modular AI application

The assignment explicitly combines Python, LangChain, OpenAI and Streamlit.

PYTHON

Deterministic financial calculations
& score logic

LANGCHAIN

Prompt templates, messages,
LLMChain + structured output

OPENAI

Chat model for educational
financial insights

STREAMLIT

Professional interactive
FinTech dashboard

CACHING

InMemoryCache + SQLiteCache
to reduce repeated calls

Modular structure: app.py • config.py • prompts.py • financial_calculator.py • chains.py • cache_manager.py • utils.py

The intelligence layer is carefully structured

Prompts carry financial context and safety rules; output follows a defined JSON schema.

01	**SystemMessage**	Educational financial-assistant role + safety
02	**HumanMessage**	Income, expenses, calculations + goal
03	**LLMChain**	Reusable chain processes the financial data
04	**JSON**	Summary • score • risk • priorities • plan

OUTPUT

Structured JSON

"financial_summary"

"financial_health_score"

"spending_analysis"

"risk_level"

"top_priorities"

"budget_recommendations"

"savings_strategy"

"next_month_action_plan"

Python calculates. AI interprets.

Deterministic metrics create a consistent numerical foundation for the LLM.

TOTAL EXPENSES

sum(all expense categories)

REMAINING INCOME

monthly income − total expenses

SAVINGS RATIO

(savings ÷ monthly income) × 100

EXPENSE RATIO

(total expenses ÷ monthly income) × 100

PRELIMINARY SCORE

82

STRONG

80–100 Strong
60–79 Generally healthy

The heuristic considers savings, leftover income, expense ratio and debt burden, with divide-by-zero protection when income is 0.

DETERMINISTIC

0–100

Faster responses, smoother interaction

Two caching modes plus streaming make the prototype practical for repeated analysis.

IN-MEMORY CACHE

Stored in RAM

Fastest • best for one session
Does not survive restart

Useful when speed matters inside a single session.

SQLITE CACHE

Stored on disk (.db)

Fast, slightly slower than RAM
Survives restart

Useful for reusing results across sessions.

STREAMING LLM .stream() → chunks → st.write_stream() LIVE ✦

Designed around clear test scenarios

The assignment defines five scenarios to check both calculations and expected AI behavior.

8,000 income / ~2,000 expenses	High remaining	High score • LOW risk
2,000 income / ~2,600 expenses	Negative remaining	Low score • HIGH risk
5,000 income / 2,500 debt	High debt share	MED/HIGH risk • debt focus
4,000 income / 1,200 savings	~30% savings ratio	High score • LOW risk
3,000 income / 3,000 expenses	Zero remaining	MED/HIGH risk • find savings room

NEXT

CSV history • visualizations • month comparison • PDF reports • goal tracker • multi-currency

FINWISE AI

Make every month more understandable.

A hybrid of financial calculation, LangChain orchestration and educational AI guidance.

ANALYSE

Turn monthly numbers into a clear picture.

EXPLAIN

Generate structured educational insights.

ACT

Leave users with priorities and a next-month plan.

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IMPORTANT

Educational prototype only.

Not financial advice. Does not execute financial transactions, connect to real bank accounts, or guarantee financial outcomes.

For real financial decisions, consult a qualified financial professional.

Thank you • Questions?